Visualizing the Impact of Corporate Investment on the U.S. Housing Market

Team 092: Tung Duong, Leonard Sanders, Jennifer Spice, Nhan Vu, Davis Cox, Nargis Sultani

1. Introduction

Corporate investors have become increasingly active in the U.S. housing market, raising concerns about affordability and local market pressures. Yet existing tools offer only fragmented views of investor activity, home values, and rent trends. This project develops an interactive, multi-scale visualization platform that integrates Zillow home value and rent time series data with Redfin's investor-share data (2019–2025), supported by city-level clustering and forecasting of housing trends. By linking these analyses within a unified map interface, the system enables users, including researchers, policymakers, and community organizations to explore where investor activity is concentrated, how prices and rents are evolving, and how future conditions may develop.

2. Problem Definition

2.1. Non-Technical Summary

Our project asks a simple question: *How does corporate investor activity shape housing affordability across the U.S.?* To assist with answering this question, we built an interactive, web-based platform that allows users to:

- identify areas with high investor activity.
- compare cities with similar multi-year patterns.
- track changes in rents and home values.
- explore trends at the state, county, ZIP-code, and city levels.
- view straightforward forecasts of future price and rent trajectories.

The platform replaces raw spreadsheets with clear, map-based visualizations, helping the public and policymakers more easily interpret how investor behavior interacts with housing pressures.

2.2. Formal Problem Definition

This project investigates how corporate investor activity relates to housing market conditions across multiple geographic zones. We analyze time-varying indicators, such as home values, rent levels, and investor market share across states, counties, cities, and ZIP-codes to identify spatial patterns and temporal dynamics linked to investor participation. The goals are to:

- combine Zillow housing time series, Redfin investor metrics, and geospatial boundaries into a unified analytical dataset.
- describe how home values and rents change over time across regions.
- measure and visualize investor activity and its geographic concentration.
- cluster cities using multi-year investor-share data to identify groups with similar trajectories.
- provide interactive, multi-scale visualizations for spatial and temporal comparison.
- produce time-series forecasts at state, county, and ZIP-code levels to project future housing trends.

The outcome is an integrated platform supporting descriptive, comparative, and predictive analysis of investor-influenced housing dynamics across the U.S.

3. Literature Survey

Corporate ownership of single-family homes expanded sharply after the 2008 foreclosure wave, reflecting a broader "financialization of housing" in which institutional investors and real estate investment trusts (REITs) treat homes as assets rather than community-stabilizing residences[1,6,10,17]. Studies show that these investors often target economically vulnerable neighborhoods, limiting homeownership access and contributing to displacement[2,4,5]. Corporate landlords typically charge higher rents, maintain stricter management practices, and exhibit higher eviction rates than small landlords, with algorithmic systems further reinforcing tenant instability[9,19,22,23]. Investor activity is also tied to gentrification and racialized displacement, while owner-occupancy tends to support neighborhood stability and wealth building[3,8,12,13,16,18,20,21].

Despite extensive research, most existing analyses remain static, limited in geographic scope, or inaccessible to the public. National housing reports rarely support ZIP-code or city-level exploration, while academic case studies offer depth but limited coverage[2,14,17]. These gaps make it difficult for communities and policymakers to evaluate localized investor impacts. Our project addresses this need by integrating Zillow and Redfin datasets into an interactive map-based platform that visualizes investor activity, home values, and rent trends. Prior housing analytics research informs our methodology[7,11,15], and the shortcomings of Zillow Offers underscore the importance of transparent, interpretable forecasting. Visualization is widely recognized as essential for democratizing access to housing data and supporting informed policy responses[24].

4. Proposed Method

4.1. Intuition

Investor activity and housing price trends operate at different spatial scales, so the system applies different methods to each. Investor behavior varies most meaningfully across cities, making Redfin's multi-year investor-share data suitable for city-level clustering. This reveals broad patterns, such as cities with rising, volatile, or stable investor activity.

Housing price trends differ across states, counties, and ZIP-codes, where Zillow's ZHVI time series provides consistent local measurements. These data support forecasting, which is integrated into the interactive map to visualize regional housing trajectories.

In essence:

- City-level clustering captures structural patterns in investor activity.
- Multi-scale forecasting models localized housing price dynamics.
- The interactive map ties the forecasting results together for intuitive exploration.

Collectively, these components form a coherent pipeline that contextualizes investor behavior while making local housing trends accessible and geographically meaningful.

4.2. Overview of the System Architecture

4.2.1 Data Sources

This project relies on housing market data obtained from two primary sources: Redfin and Zillow. Redfin provides annual city-level measures of investor home purchases and investor market share for the years 2019 through 2025. Zillow contributes to the ZHVI home value time-series, available at state, county, and ZIP-code levels over the same period. Together, these sources supply both the investor activity information and the home value trends needed for clustering and forecasting.

Geographic visualization is enabled through the Census Bureau's cartographic boundary shapefiles and helper code from the U.S. Atlas TopoJSON[25] project. ZIP-code to county references were sourced from the U.S. Department of Housing and Urban Development[26].

4.2.2 Data Processing

Data preparation emphasized geographic consistency and continuity across the years. City names and geographic identifiers were standardized, and locations lacking complete time-series histories were removed. Investor variables were normalized to support clustering, while ZHVI sequences were reformatted into forecasting-ready inputs. The cleaned datasets were stored and managed in PostgreSQL, which was used to organize into hierarchical state/county/ZIP-code structures to enable smooth navigation in the interactive map. These steps ensured reliable analytical output across all components of the project.

4.2.3 Clustering of Investor Activity (City Level)

Clustering was performed using multi-year investor trends to identify groups of cities with similar patterns of corporate purchasing. Each city was represented by its annual investor purchase counts and investor share over seven years. After standardizing these variables, a distance-based clustering model was applied, with the number of clusters selected using quantitative validation. The resulting groups highlight distinct investor dynamics, such as consistently high activity, rising markets, or declining involvement. This clustering analysis functions as an independent analytical layer that characterizes investor behavior nationwide; its detailed results are presented later in the evaluation section.

4.2.4 Forecasting Housing Prices

A Holt-Winters exponential smoothing time-series model was performed to forecast investor market share and median sale price trends across multiple geographic dimensions in the United States, including states, counties, and ZIP-codes. Each dimension was grouped into quarterly or annual time series and modeled using exponential smoothing. This model type allows the algorithm to learn level and trend components while accounting for data variability across different geographies. Separate models were fit for each ZIP-code, county, and state, with holdout periods used to evaluate error and to ensure that forecasts extended beyond the most recent actuals. The model results include both performance metrics and future projections used to identify directional changes in investor activity and home pricing.

By generating geographically granular predictions for investor market share and state level median sale prices, the exponential smoothing models provide insights into how investor presence and housing costs may change over the coming quarters and years. The model outputs capture variation across different geographical dimensions where some exhibited linear increase over time, and some stayed the same or even decreased. The model results also allow for detailed comparisons across dimensions which can be valuable for a buyer figuring out where to purchase a home. These detailed forecasts are compiled in the interactive map created in the project and are shown via a hover over tool tip to assist home buyers.

4.2.5 Interactive Map Visualization

The interactive map, implemented with **D3.js**, provides the primary interface for exploring housing trends. It supports multi-scale navigation across state, county, and ZIP-code geographies and presents current home values, historical trends, and forecasted trajectories through choropleth shading and interactive panels. Users may click regions to view detailed time-series plots and metadata. The map integrates all forecasting outputs and enables intuitive geographic exploration of housing dynamics. Clustering results are not incorporated into the map; they remain part of the independent analytical evaluation.

5. Evaluation

This section presents the analytical results of the project across its three components: clustering of investor activity at the city level, forecasting of home values at state, county and ZIP-code sub-levels, and evaluation of the interactive geographic visualization. Each component offers a complementary perspective: clustering reveals patterns in investor behavior, forecasting provides forward-looking estimates of market trends, and the map integrates these insights into an exploratory interface.

5.1 Clustering Results (City-Level Analysis)

The clustering analysis identifies groups of U.S. cities that exhibit similar patterns in investor activity between 2019 and 2025. Using Redfin's annual investor-share data, cities were clustered based on trajectories rather than static values, allowing the method to capture both magnitude and trend similarities. This enables comparisons of how investor presence has evolved across metropolitan areas over time.

Several distinct temporal patterns emerge. Some cities display persistently high investor penetration; others experience a sharp surge following 2020, while a smaller group shows notable declines after peaking mid-period. These patterns provide insight into how institutional investment has spread, and in some cases retreated, across different local markets.

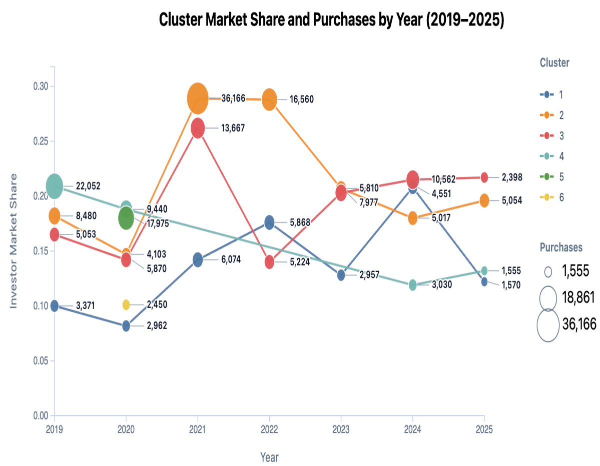

Figure 1. Market Share by Cluster

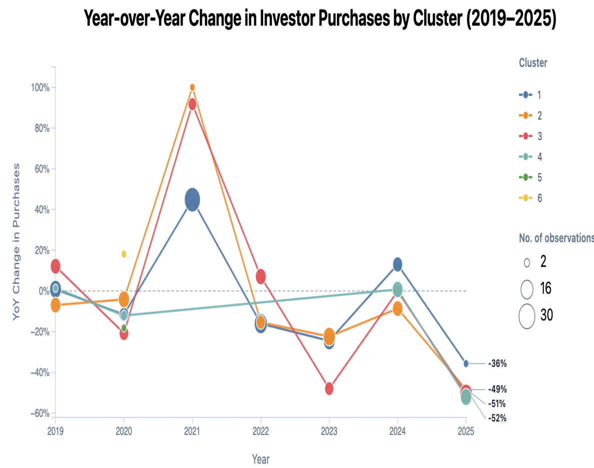

Figure 2. Change in Investor Purchases by Cluster

To illustrate the evolution of investor share over time, annual grids were produced that display each city's investor share for every year in the dataset. This clustering analysis is an independent analytical component. It is not connected to the interactive map but provides essential context for understanding higher-level patterns in corporate investor behavior nationwide.

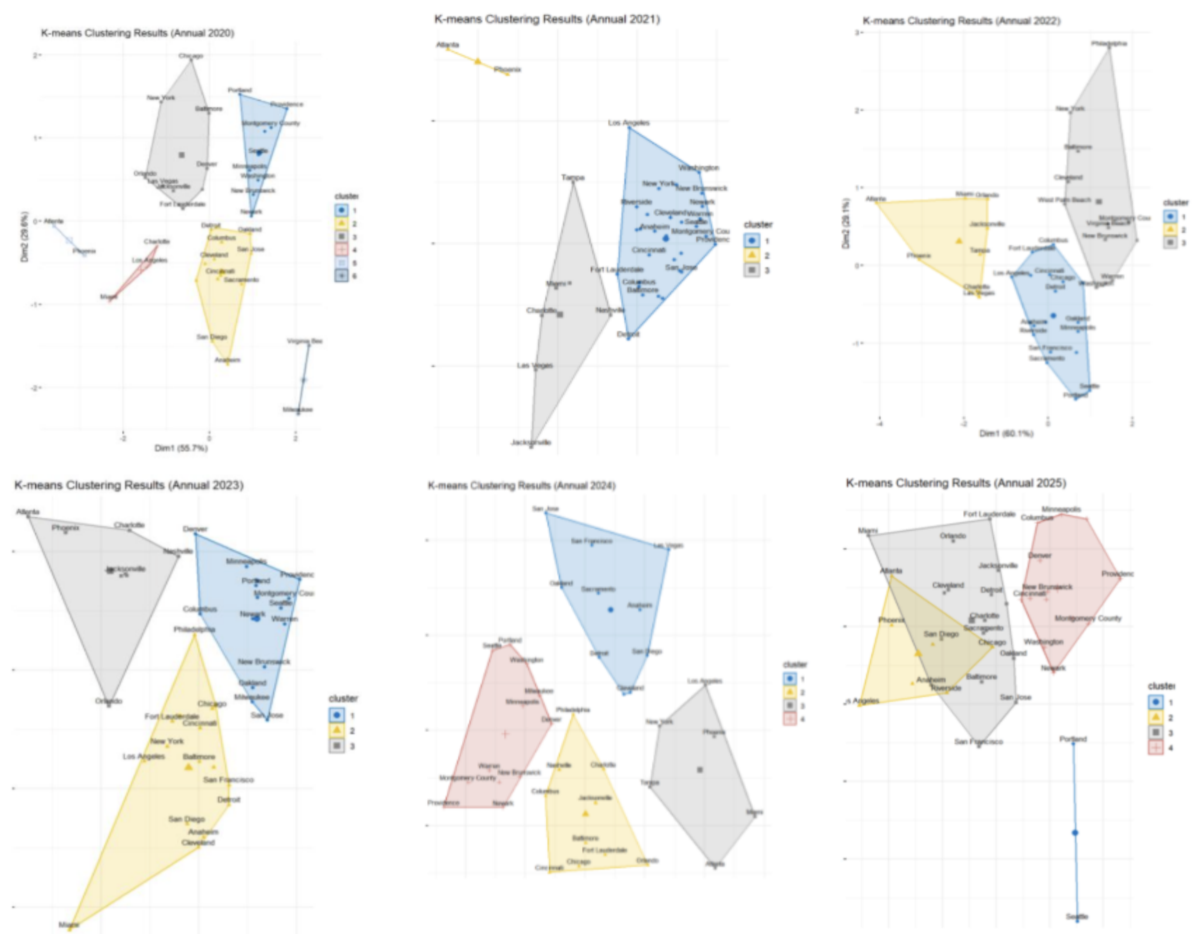

Figure 3. Clustering Analysis Illustrating Corporate Investor Behavior

5.2 Forecasting Results (State, County, ZIP-code Levels)

Forecasting models were applied to Zillow home-value time series (ZHVI) across state, county, and ZIP-code geographic levels. The goal is to identify differences in expected future price trajectories across regions with varying levels of investor activity. These forecasts highlight areas experiencing elevated appreciation, stagnation, or volatility. The forecasting component directly supports the interactive visualization system. Predicted curves provide users with forward-looking insights that complement historical patterns displayed on the map.

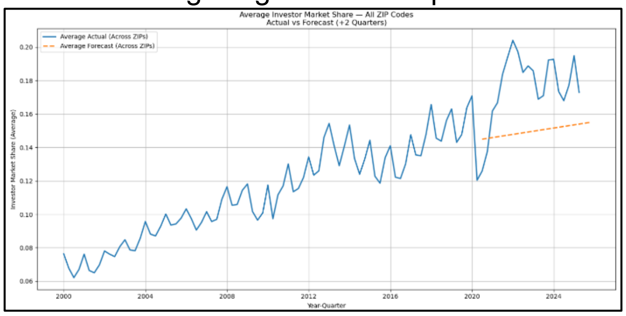

Figure 4. Average Investor Market Share for All Zip Codes

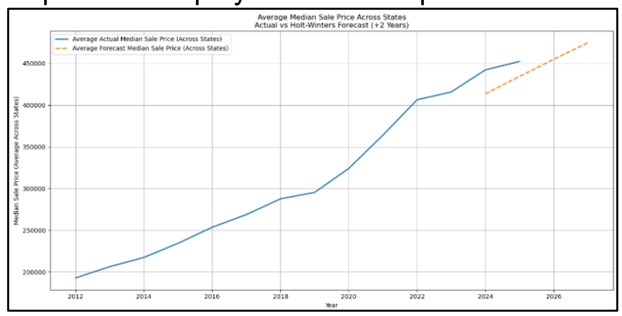

Figure 5. Average Median Sales Price Across All States

5.3 Interactive Map Evaluation

The interactive, multi-scale map integrates investor-share data with historical and forecasted home-value trends. It enables users to explore conditions at the state, county, and ZIP-code levels, revealing geographic variation in market exposure to investor activity.
Key features evaluated include:

- Stability and responsiveness during multi-scale navigation
- Legibility of investor-share patterns
- Accuracy of mapped geographic boundaries
- Alignment between time-series forecasts and map-selected regions

The map's ability to visually compare investor activity with current and projected home values enhances interpretability and provides a coherent environment for exploratory analysis.

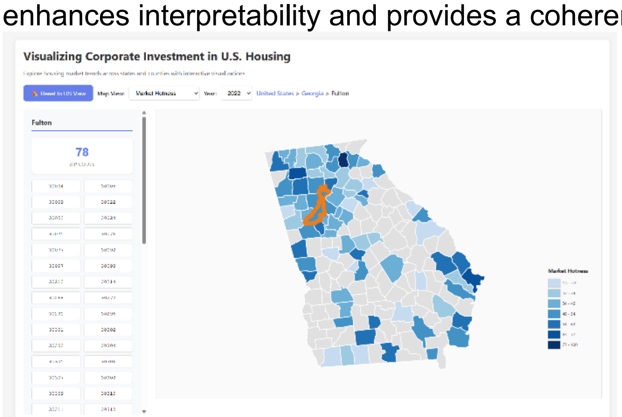

Figure 6. State of Georgia, with Fulton County selected

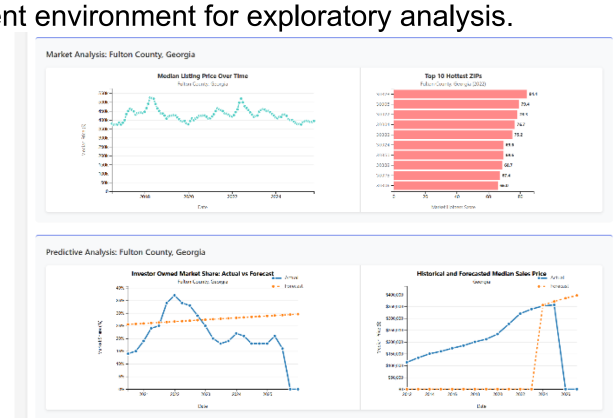

Figure 7. Market and Predictive Analysis for Fulton County, Georgia

6. Conclusions and Discussion

This project provides an integrated framework for examining how corporate investors influence U.S. housing markets by combining clustering analysis, multi-scale forecasting, and interactive geographic visualization. The results show that investor activity varies substantially across cities, forming distinct behavioral groups rather than a uniform national pattern. This highlights the geographically uneven nature of institutional investment.

The forecasting models extend the analysis by projecting home-value trends at state, county, and ZIP-code levels. These projections show that regions with high investor presence often exhibit different expected price trajectories, illustrating potential future implications for affordability. Integrating these forecasts into the interactive map enables accessible, multi-scale exploration of localized housing dynamics.

The interactive map serves as the project's primary user-facing tool, translating complex datasets into an intuitive interface that supports comparative analysis across geographic scales.

6.1 Limitations

The clustering analysis is limited to cities with complete investor time-series data, and forecasting accuracy depends on broader economic conditions outside the model's scope. The interactive map currently integrates only the forecasting component, leaving clustering as a separate analytical result.

Furthermore, the primary purpose of ZIP-code is to facilitate efficient mail sorting and delivery, and therefore does not have a one-to-one relationship with county. This has the less than desirable effect of conflating data between multiple counties where ZIP-code does cross county lines. However, as ZIP-code is an intuitive demarcation for users, it's use in the analysis is beneficial.

6.2 Future Work

Future extensions may incorporate demographic variables, rental data, scenario-based forecasting, property tax information, or integration of clustering outputs into the map to provide a more unified tool for understanding investor impacts. As an increasing number of neighborhoods subject to home-owner associations prevent non-owner-occupied housing, an analysis of such neighborhoods compared to those permitting non-owner-occupied housing may prove insightful.

6.3 Team member Contribution

All team members have contributed a similar amount of effort.

References

1. Aalbers, M. B. (2015). The variegated financialization of housing. *International Journal of Urban and Regional Research*, 39(4), 542–554. https://doi.org/10.1111/1468-2427.12216
2. An, B. Y. (2023). The Influence of Institutional Single-Family Rental Investors on Homeownership: Who Gets Targeted and Pushed Out of the Local Market? *Journal of Planning Education and Research*, *44*(4), 2231-2250. https://doi.org/10.1177/0739456X231176072 (Original work published 2024)
3. Been, V., Ellen, I. G., & O'Regan, K. (2019). Supply skepticism: Housing supply and affordability. *Housing Policy Debate*, 29(1), 25–40. https://doi.org/10.1080/10511482.2018.1476899
4. Charles, S. L. (2019). The financialization of single-family rental housing: An examination of real estate investment trusts' ownership of single-family houses in the Atlanta metropolitan area. *Journal of Urban Affairs, 41*(7), 1031–1046. https://doi.org/10.1080/07352166.2019.1662728
5. Christophers, B. (2021). How and Why U.S. Single-Family Housing Became an Investor Asset Class. *Journal of Urban History*, *49*(2), 430-449. https://doi.org/10.1177/00961442211029601 (Original work published 2023)
6. Colburn, G., Walter, R. J., & Pfeiffer, D. (2020). Capitalizing on Collapse: An Analysis of Institutional Single-Family Rental Investors. *Urban Affairs Review*, *57*(6), 1590-1625. https://doi.org/10.1177/1078087420922910 (Original work published 2021)
7. DiNauta, M. & Sun, J. (2023, February 13). *How Zillow data science measures business outcomes with Bayesian statistics*. Zillow. https://www.zillow.com/tech/how-zillow-data-science-measures-business-outcomes-with-bayesian-statistics/
8. DiPasquale, D., & Glaeser, E. L. (1999). Incentives and social capital: Are homeowners better citizens? *Journal of Urban Economics*, 45(2), 354–384. https://doi.org/10.1006/juec.1998.2098
9. Fesko, Luke, Landlord Corporatization and Eviction in North Carolina. http://dx.doi.org/10.2139/ssrn.5418375
10. Fields, D., & Uffer, S. (2016). The financialisation of rental housing: A comparative analysis of New York City and Berlin. *Urban Geography*, 37(4), 546–567. https://doi.org/10.1080/02723638.2015.1069020
11. Fu, Runshan and Huang, Yan and Mehta, Nitin and Singh, Param Vir and Srinivasan, Kannan, Unequal Impact of Zestimate on the Housing Market (June 15, 2023). http://dx.doi.org/10.2139/ssrn.4480469
12. Garriga, C., Gete, P., & Tsouderou, A. (2023). The economic effects of real estate investors. *Real Estate Economics*, 51, 655–685. https://doi.org/10.1111/1540-6229.12427
13. Goodman, J. (2003). Homeownership and Investment in Real Estate Stocks. *The Journal of Real Estate Portfolio Management*, *9*(2), 93–106. http://www.jstor.org/stable/24882313
14. Hermann, A. (2023, July 18). *8 facts about investor activity in the single-family rental market*. Joint Center for Housing Studies of Harvard University. https://www.jchs.harvard.edu/blog/8-facts-about-investor-activity-single-family-rental-market
15. Kiger, P. J. (2021, December 9). *Flip flop: Why Zillow's algorithmic home buying venture imploded.* Stanford Graduate School of Business.
16. Lee, K., & Wylie, D. (2024). *Institutional investors, rents, and neighborhood change in the single family residential market* (Federal Reserve Bank of Philadelphia Working Paper No. 24-13). https://doi.org/10.21799/frbp.wp.2024.13
17. Mills, J., Molloy, R. and Zarutskie, R. (2019), Large-Scale Buy-to-Rent Investors in the Single-Family Housing Market: The Emergence of a New Asset Class. Real Estate Economics, 47: 399-430. https://doi.org/10.1111/1540-6229.12189

18. Newman, K., & Wyly, E. K. (2006). The right to stay put, revisited: Gentrification and resistance to displacement in New York City. *Urban Studies*, 43(1), 23–57. https://doi.org/10.1080/00420980500409385
19. Raymond, E., Duckworth, R., Miller, B., Lucas, M., & Pokharel, S. (2018). From foreclosure to eviction: Housing insecurity in corporate-owned single-family rentals. *Housing Studies*, 33(8), 1102–1129. https://doi.org/10.1080/02673037.2018.1447090
20. Raymond, E. L., Miller, B., McKinney, M., & Braun, J. (2021). Gentrifying Atlanta: Investor Purchases of Rental Housing, Evictions, and the Displacement of Black Residents. *Housing Policy Debate*,B*31*(3–5), 818–834. https://doi.org/10.1080/10511482.2021.1887318
21. Rohe, W. M., & Stewart, L. S. (1996). Homeownership and neighborhood stability. *Housing Policy Debate*, 7(1), 37–81. https://doi.org/10.1080/10511482.1996.9521213
22. Rosen, E., & Garboden, P. M. E. (2019). Landlord decision-making and the eviction process. *Social Service Review*, 93(1), 1–29. https://doi.org/10.1086/701836
23. Seymour, E., & Akers, J. (2021). Portfolio landlords and eviction dynamics in Atlanta and Detroit. *Housing Policy Debate*, 31(2), 316–333. https://doi.org/10.1080/10511482.2020.1812693
24. Zuk, M., Bierbaum, A. H., Chapple, K., Gorska, K., & Loukaitou-Sideris, A. (2018). Gentrification, displacement, and the role of public investment: A literature review. *Urban Studies*, 55(8), 1565–1579. https://doi.org/10.1177/0042098017710143
25. U.S. Atlas TopoJSON helper files: https://github.com/topojson/us-atlas
26. U.S. Department of Housing and Urban Development USPS Zip Code Crosswalk Files: https://www.huduser.gov/portal/datasets/usps_crosswalk.html

Heilmeier Questions

1. What are you trying to do? Articulate your objectives using absolutely no jargon.
2. How is it done today; what are the limits of current practice?
3. What's new in your approach? Why will it be successful?
4. Who cares?
5. If you're successful, what difference and impact will it make, and how do you measure them (e.g., via user studies, experiments, ground truth data, etc.)?
6. What are the risks and payoffs?
7. How much will it cost?
8. How long will it take?
9. What are the midterm and final "exams" to check for success? How will progress be measured?